

Hiring Assignment - SDE Interns

SDE Intern Assignment: EMR Feature Implementation

Thank you for applying for the internship position at SwasthiQ.

You are required to implement a simplified **Pharmacy Module** consisting of three core pages, following the design references shared.

Assignment Overview

You must build:

1. **Dashboard Page (Sales Overview)**
2. **Inventory Page**

All three pages must be backed by **fully functional REST APIs built in Python** and consumed via a **React frontend**. Backend APIs must be functional and properly integrated.

Technical Requirements

Backend Requirements

- Language: **Python**
- Framework: FastAPI or Flask (FastAPI preferred)
- Database: PostgreSQL / SQLite
- RESTful API structure mandatory
- Clear API contracts
- Proper request and response models
- Validation and error handling

API Expectations

You must implement APIs for:

Dashboard

- Get today's sales summary
- Get total items sold
- Get low stock items
- Get purchase order summary

Inventory

- List medicines
- Add new medicine
- Update medicine
- Mark expired / out of stock
- Filter and search

All endpoints must:

- Follow REST conventions
- Return proper HTTP status codes
- Use structured JSON responses

Frontend Requirements

- Framework: React (functional components + hooks)
- Must match design philosophy shown in the reference images
- Clean UI hierarchy
- Proper state management
- API integration must be real (no mock data)
- Proper loading and error states

Design expectations:

- Maintain spacing, layout balance, and component hierarchy
- Avoid unnecessary styling deviations
- Follow consistent color and typography usage
- Reusable components preferred

Pages to Implement

1. Dashboard Page

Must include:

- Sales summary card
- Items sold card
- Low stock indicator
- Purchase order summary
- Recent sales list

All data must be fetched from backend APIs.

2. Inventory Page

Must include:

- Inventory overview summary
- Complete inventory table
- Medicine details
- Status indicators (Active, Low Stock, Expired, Out of Stock)
- Add / Update functionality

Must support filtering and searching.

Submission Guidelines

1. **Single Repository:** Provide a single public GitHub repository containing the complete project. The repository must clearly separate:

- /backend (Python REST API implementation)
- /frontend (React application)
- README.md (Documentation and API contracts)

2. **Live Link:** A working, publicly hosted link (e.g., Vercel, Netlify) to the application.

3. **Technical Explanation:** A brief README explaining the **REST API structure** you designed for the functions and how your Python functions ensure data consistency upon update.

4. The Assignment is to be completed in 3 days from the date it is shared. Late submissions will not be considered.

Reference Images (Assignment to be made as per these images only)

